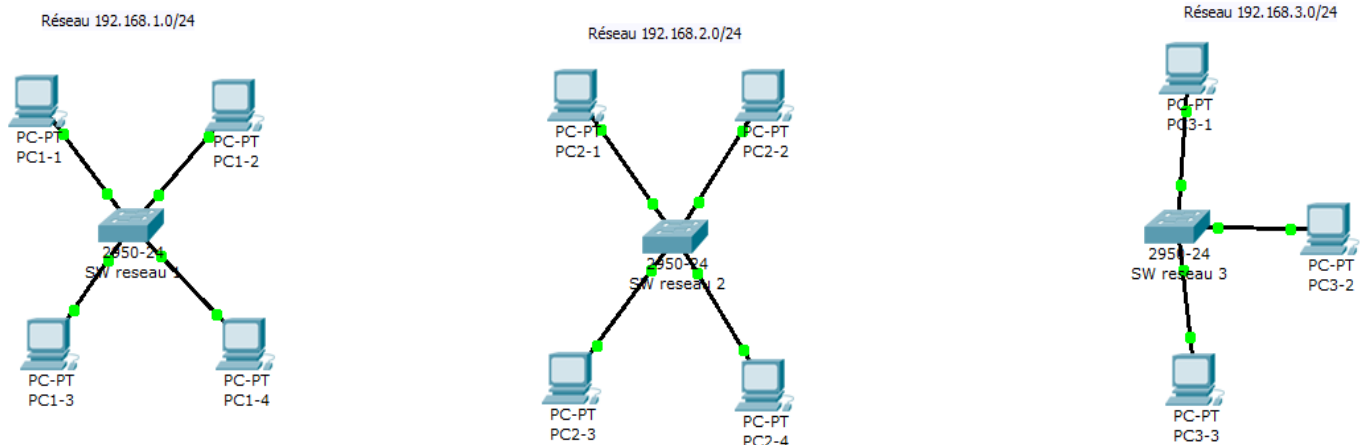
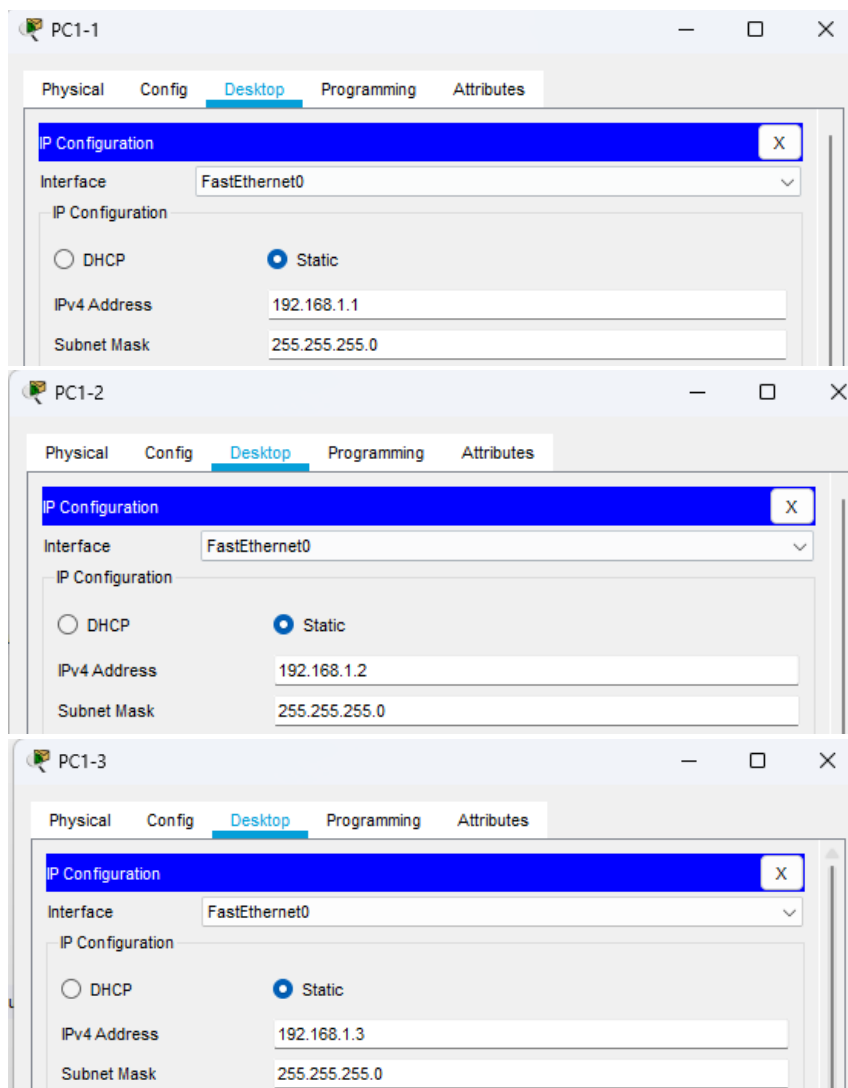


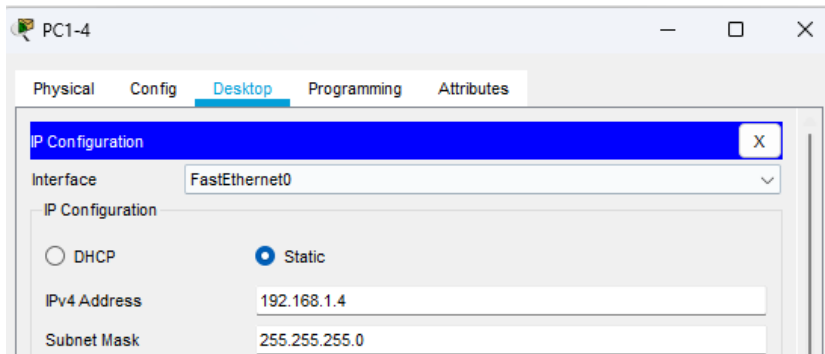
Mettre en place le réseau suivant :



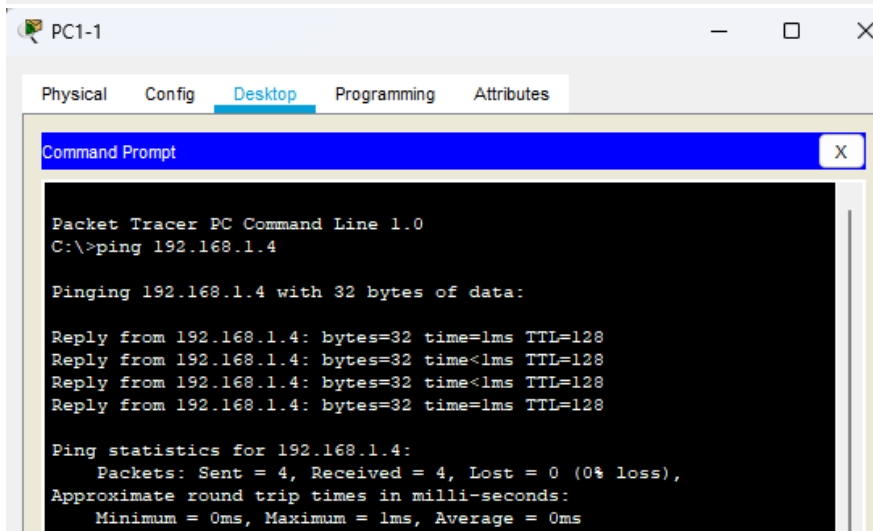
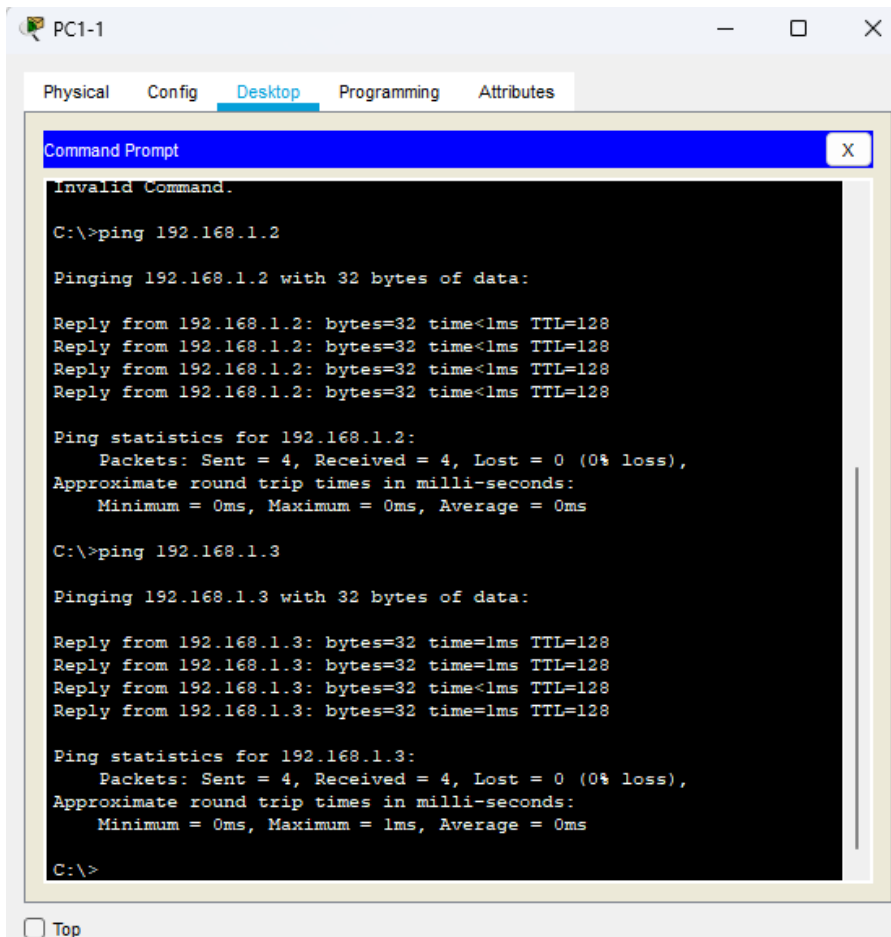
Configuration IP

Configurez les adresses IP des postes du premier réseau : vous définirez les adresses IP dans l'ordre de numérotation des PC.





Vérifiez le bon fonctionnement du réseau par des ping entre tous les postes



PC1-2

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC1-3

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

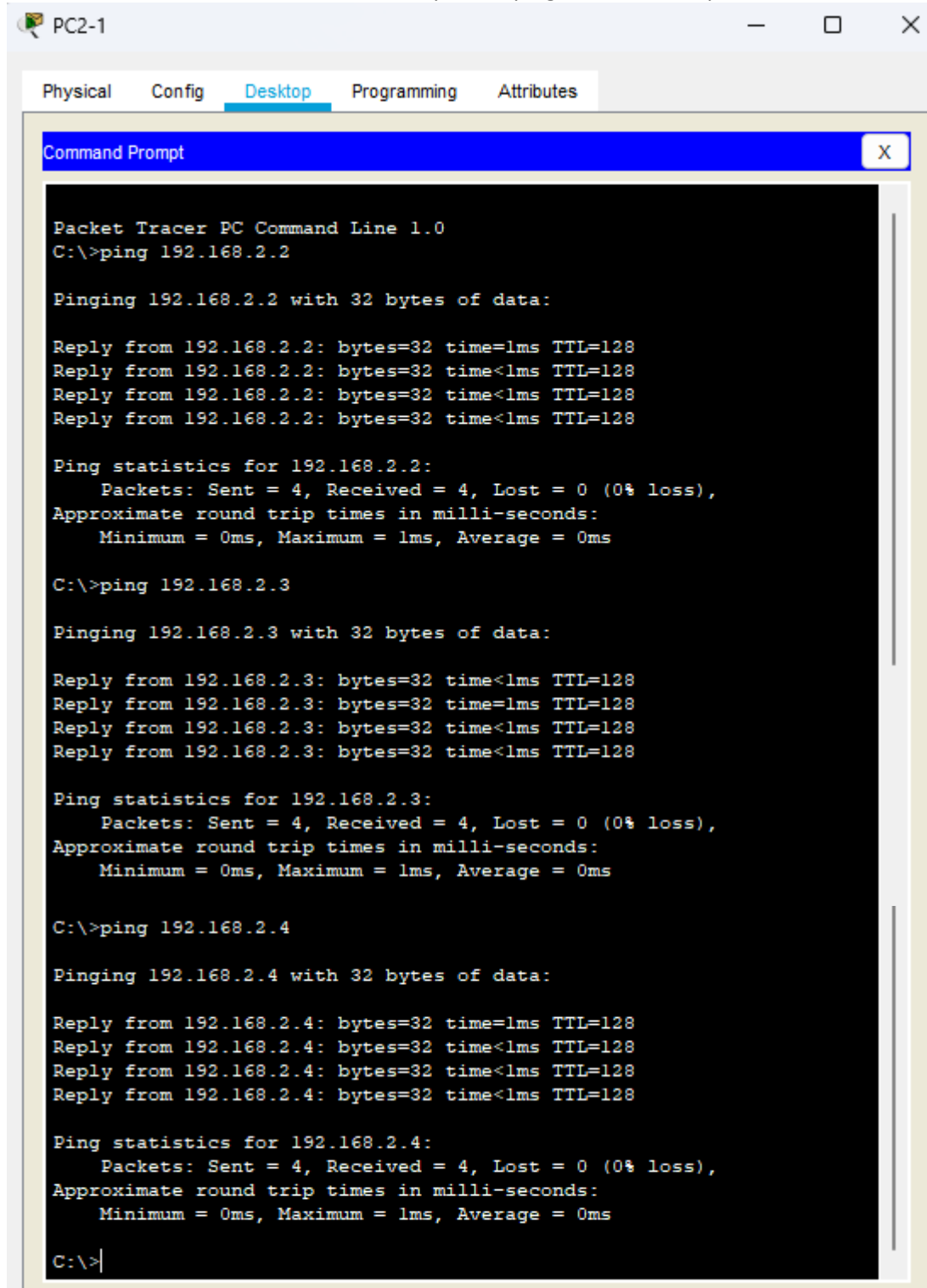
C:\>
```

Configurez les adresses IP des postes du deuxième réseau : vous définirez les adresses IP dans l'ordre de numérotation des PC.

The image displays four sequential screenshots of the 'IP Configuration' window for four different PCs, labeled PC2-1 through PC2-4. Each window has a tabbed interface with 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active in all. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The 'IPv4 Address' and 'Subnet Mask' fields are populated with the following values:

PC	IPv4 Address	Subnet Mask
PC2-1	192.168.2.1	255.255.255.0
PC2-2	192.168.2.2	255.255.255.0
PC2-3	192.168.2.3	255.255.255.0
PC2-4	192.168.2.4	255.255.255.0

Vérifiez le bon fonctionnement du réseau par des ping entre tous les postes



The screenshot shows a Packet Tracer interface for PC2-1. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The window title is 'Command Prompt' with a close button (X). The text inside the window is as follows:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

PC2-2

Physical Config Desktop Programming Attributes

Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=1ms TTL=128
Reply from 192.168.2.4: bytes=32 time=1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC2-3

Physical Config Desktop Programming Attributes

Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.4

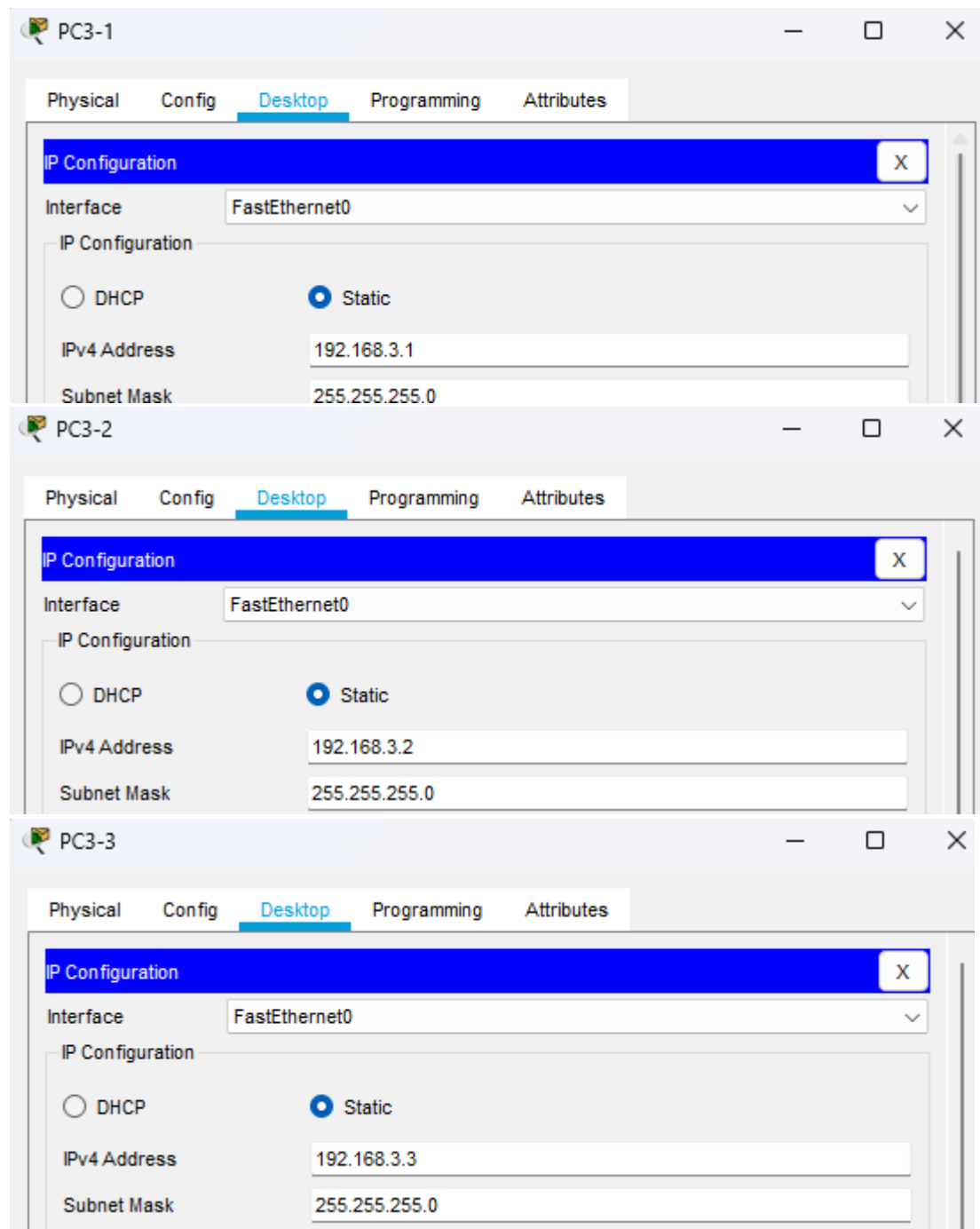
Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

Configurez les adresses IP des postes du troisième réseau : vous définirez les adresses IP dans l'ordre de numérotation des PC.



Vérifiez le bon fonctionnement du réseau par des ping entre tous les postes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

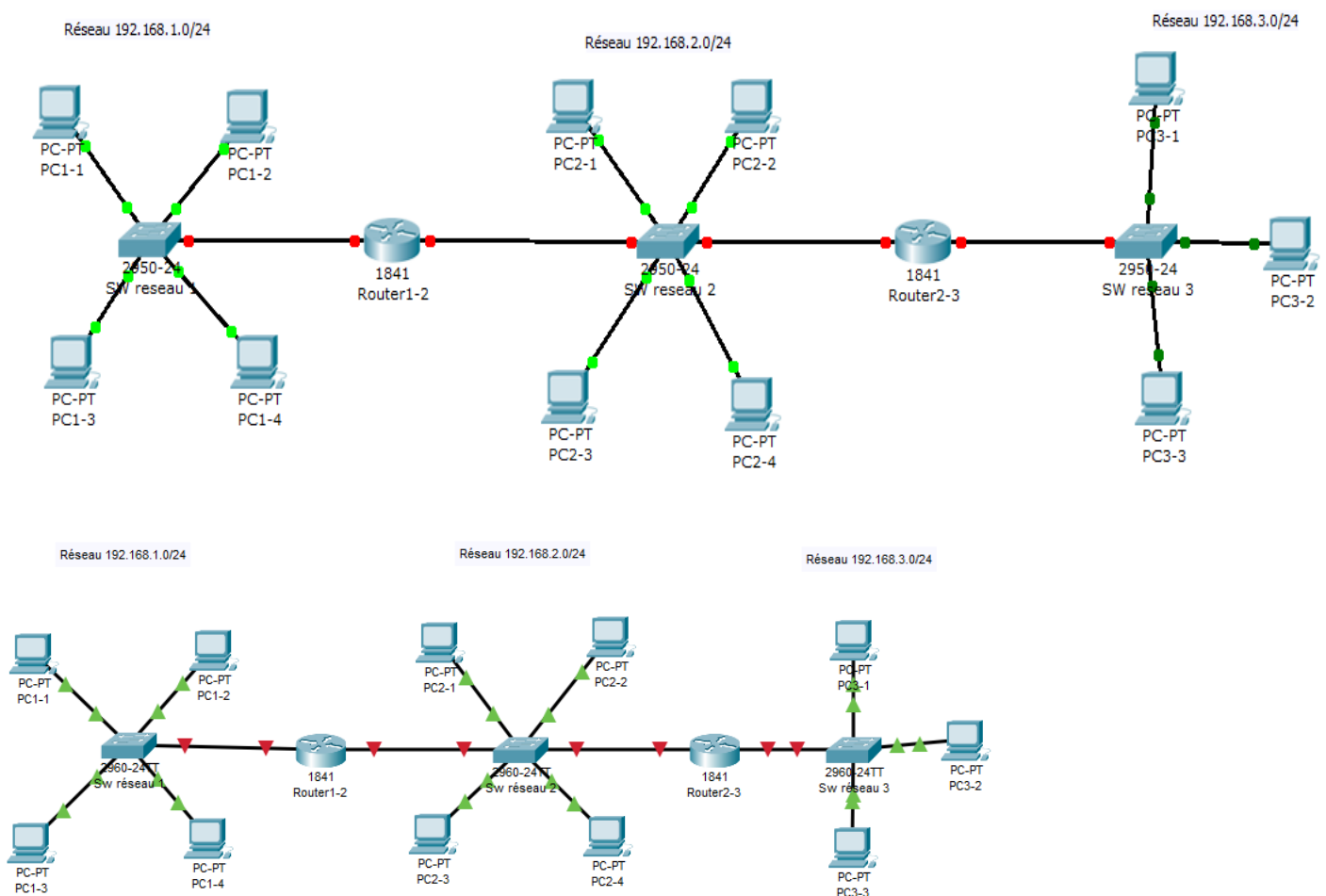
Reply from 192.168.3.3: bytes=32 time=1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```


Mise en place de routeurs

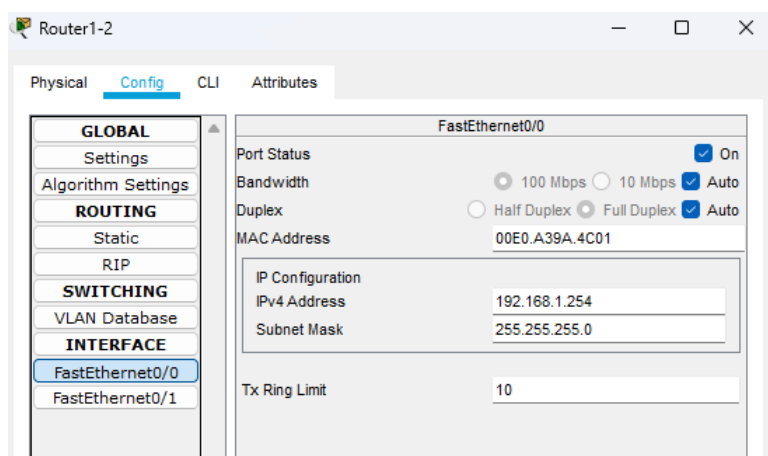
Vous devez obtenir le schéma suivant :



Configuration des routeurs

Routeur1-2

Interface sur le réseau 1 : dernière adresse du réseau.



Interface sur le réseau 2 : avant dernière adresse du réseau.

The screenshot shows the configuration window for Router1-2. The 'Config' tab is active. On the left, the 'INTERFACE' section is expanded, and 'FastEthernet0/1' is selected. The main area displays the configuration for FastEthernet0/1. The 'Port Status' is 'On'. 'Bandwidth' is set to '100 Mbps' and 'Auto'. 'Duplex' is set to 'Full Duplex' and 'Auto'. The 'MAC Address' is '00E0.A39A.4C02'. The 'IP Configuration' section shows 'IPv4 Address' as '192.168.2.253' and 'Subnet Mask' as '255.255.255.0'. The 'Tx Ring Limit' is '10'.

Routeur2-3

Interface sur le réseau 2 : dernière adresse du réseau.

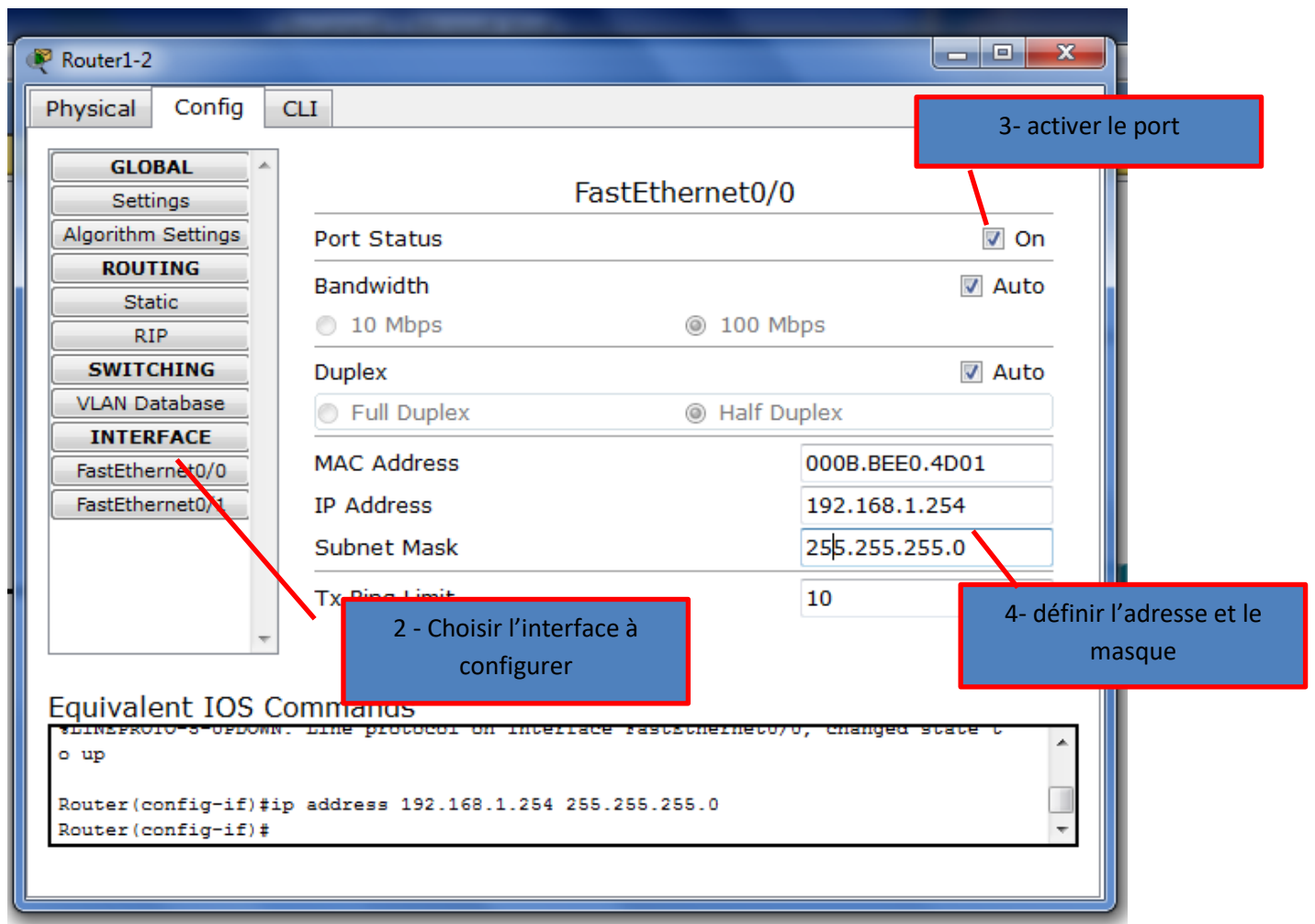
The screenshot shows the configuration window for Router2-3. The 'Config' tab is active. On the left, the 'INTERFACE' section is expanded, and 'FastEthernet0/0' is selected. The main area displays the configuration for FastEthernet0/0. The 'Port Status' is 'On'. 'Bandwidth' is set to '100 Mbps' and 'Auto'. 'Duplex' is set to 'Full Duplex' and 'Auto'. The 'MAC Address' is '0001.641C.7401'. The 'IP Configuration' section shows 'IPv4 Address' as '192.168.2.254' and 'Subnet Mask' as '255.255.255.0'. The 'Tx Ring Limit' is '10'.

Interface sur le réseau 3 : avant dernière adresse du réseau.

The screenshot shows the configuration window for Router2-3. The 'Config' tab is active. On the left, the 'INTERFACE' section is expanded, and 'FastEthernet0/1' is selected. The main area displays the configuration for FastEthernet0/1. The 'Port Status' is 'On'. 'Bandwidth' is set to '100 Mbps' and 'Auto'. 'Duplex' is set to 'Full Duplex' and 'Auto'. The 'MAC Address' is '0001.641C.7402'. The 'IP Configuration' section shows 'IPv4 Address' as '192.168.3.253' and 'Subnet Mask' as '255.255.255.0'. The 'Tx Ring Limit' is '10'.

IHM de configuration d'une interface de routeur :

1- Onglet à sélectionner

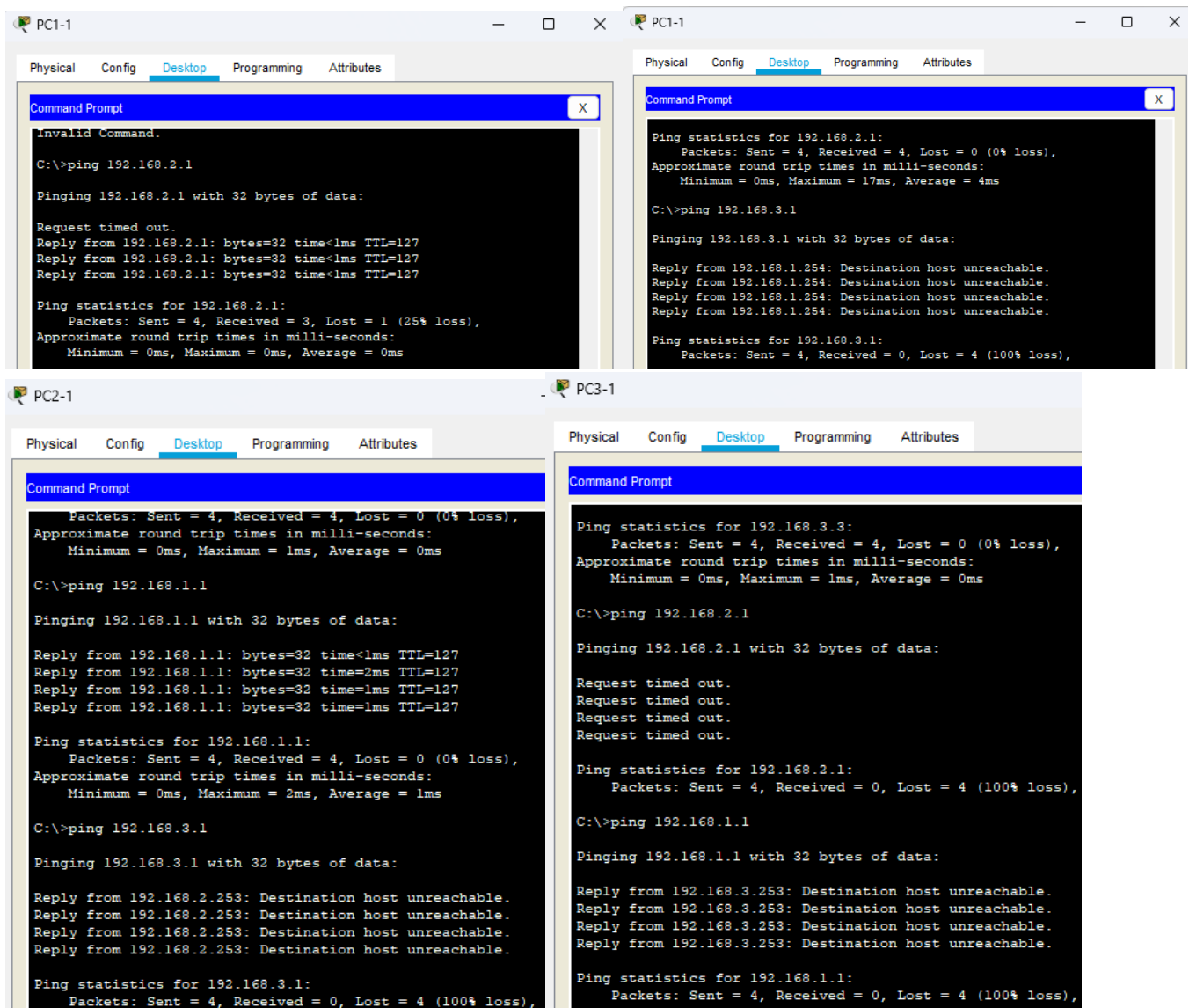


NOTA :

Pour connaître le nom d'une interface du routeur (FastEthernet0/0 alias Fa0/0 ou FastEthernet0/1 alias Fa0/1), approcher le pointeur de la souris sur le câble qui est branché sur l'interface.

Configurer les passerelles : maintenant qu'il y a des routeurs on peut indiquer sur chaque poste quelle est sa passerelle.

Tester par des ping



Écrire une petite note expliquant quelles sont les communications qui fonctionnent et celles qui ne fonctionnent pas.

Les pings entre les réseaux 1-2 fonctionnent mais pas entre le réseau 2-3 et 1-3 car les machines du réseau 1 et 2 sont configurées pour avoir routeur 1-2 comme passerelle qui ne connaît pas le réseau 3. Pour que les pings entre les réseaux 1-3 et 2-3 fonctionnent, il faut mettre des routes statiques pour indiquer au routeur 1-2 que le réseau 3 est accessible via le routeur 2-3 qui a comme adresse IP 192.168.2.254 et au routeur 2-3 que le réseau 1 est accessible via le routeur 1-2 qui a comme adresse IP 192.168.2.253.

Ajouter les routes manquantes.

Pour le Routeur 1-2 :

ip route 192.168.3.0 255.255.255.0 192.168.2.254

Donc "Pour accéder au réseau 192.168.3.0/24 envoie la trame au routeur 192.168.2.254"

Pour le Routeur 2-3 :

ip route 192.168.1.0 255.255.255.0 192.168.2.253

Donc "Pour accéder au réseau 192.168.1.0/24 envoie la trame au routeur 192.168.2.253"

PC1-1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=126
Reply from 192.168.3.1: bytes=32 time=11ms TTL=126
Reply from 192.168.3.1: bytes=32 time<1ms TTL=126
Reply from 192.168.3.1: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms
```

PC2-2

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.1: bytes=32 time=11ms TTL=127
Reply from 192.168.3.1: bytes=32 time=11ms TTL=127
Reply from 192.168.3.1: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 7ms
```

PC3-1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.1: bytes=32 time=12ms TTL=126
Reply from 192.168.1.1: bytes=32 time=14ms TTL=126
Reply from 192.168.1.1: bytes=32 time=17ms TTL=126

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 17ms, Average = 14ms
```